QP CODE:25900083

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCFTQ100 - BASIC BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap) Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any 5 questions Each question carries 2 marks

1. **With the help of structure, list the properties of insoluble component of starch [A] / [CO1]**

   The water-insoluble component of starch is primarily amylopectin. It's a branched, highly branched polysaccharide that forms a complex network within the starch granule. This structure contributes to the granule's semi-crystalline nature and its resistance to easy dissolution in cold water.

   Draw the structure- 1 mark

2. **What does the secondary structure of proteins refer to? [U] / [CO2]**

   The secondary structure of a protein refers to the local folding patterns of the polypeptide backbone, specifically the shapes formed by hydrogen bonds between amino acids in the chain. The most common secondary structures are the **α-helix and the β-pleated sheet.**

3. **What is halogenation of fats? Discuss its importance. [A] / [CO4]**

   Halogenation of fats involves introducing halogen atoms (fluorine, chlorine, bromine, or iodine) into a fatty acid molecule. This can occur naturally, where halogenated fatty acids are formed in various organisms, or through chemical reactions. In chemical reactions, halogens can be added to the double bonds of unsaturated fatty acids, or they can substitute hydrogen atoms on the carbon chain.

   Halogenation of fats, the introduction of halogen atoms (like chlorine or bromine) into fatty acids, offers several advantages, including enhanced biological activity, use as analytical tools, and the creation of new materials. Halogenated fatty acids can exhibit antibacterial, antifungal, and anticancer properties, and they can be used to develop more effective drugs. They are also valuable in analytical chemistry, especially in determining

iodine values (IV). Furthermore, halogenation can be a starting point for producing various other materials.

4. **Define and give the importance of Kiliani-Fischer synthesis [U] / [CO1]**
   The Kiliani-Fischer synthesis is a method used to lengthen the carbon chain of aldoses (sugars) by adding one carbon atom. It involves reacting an aldose with hydrogen cyanide (HCN) to form a cyanohydrin, which is then reduced to an aldehyde and hydrolyzed to yield the longer chain aldose. This process creates two epimers (diastereomers) at the new stereogenic center.
5. **How does enzyme perform its catalytic role? [U] / [CO3]**
   Enzymes accelerate reactions by lowering the activation energy required for the reaction to occur. They achieve this by binding to the substrate, orienting it properly for the reaction, and sometimes directly participating in the chemical transformation. This process involves several steps: substrate binding, activation of the substrate, stabilization of the transition state, and product release.
6. **Classify fatty acids based on their importance in diet. [An] / [CO4]**
   Fatty acids can be classified based on their dietary importance into essential and non-essential. Essential fatty acids, like omega-3 and omega-6, must be obtained from the diet because the body cannot synthesize them. Non-essential fatty acids, like saturated and monounsaturated fats, can be produced by the body.
7. **What are Ligases? Give examples. [K] / [CO3]**

Ligases are enzymes that catalyze the formation of chemical bonds between two molecules, typically with the help of ATP (adenosine triphosphate). They are essential for processes like DNA replication, repair, and protein synthesis.

Examples : Triose –phosphate isomerase Ligases catalyze the joining of two molecules by new covalent bonds with concomitant hydrolysis of Adenosine-triphosphate (ATP) or other similar molecules. Example: glutamine synthetase. Translocases catalyze the movement of ions or molecules across cell membranes in general.

[2x5 = 10]

Part B

Short Essay Questions Answer any 5 questions Each question carries 4 marks

8. **Make a detailed note on the following [U] / [CO1]**
   (1 marks each)

a. Mutarotation- Mutarotation is the change in optical rotation of a solution of a cyclic sugar (like glucose) due to the interconversion of its anomeric forms (alpha and beta) when dissolved in water. This phenomenon occurs because the alpha and beta anomers interconvert until they reach an equilibrium.

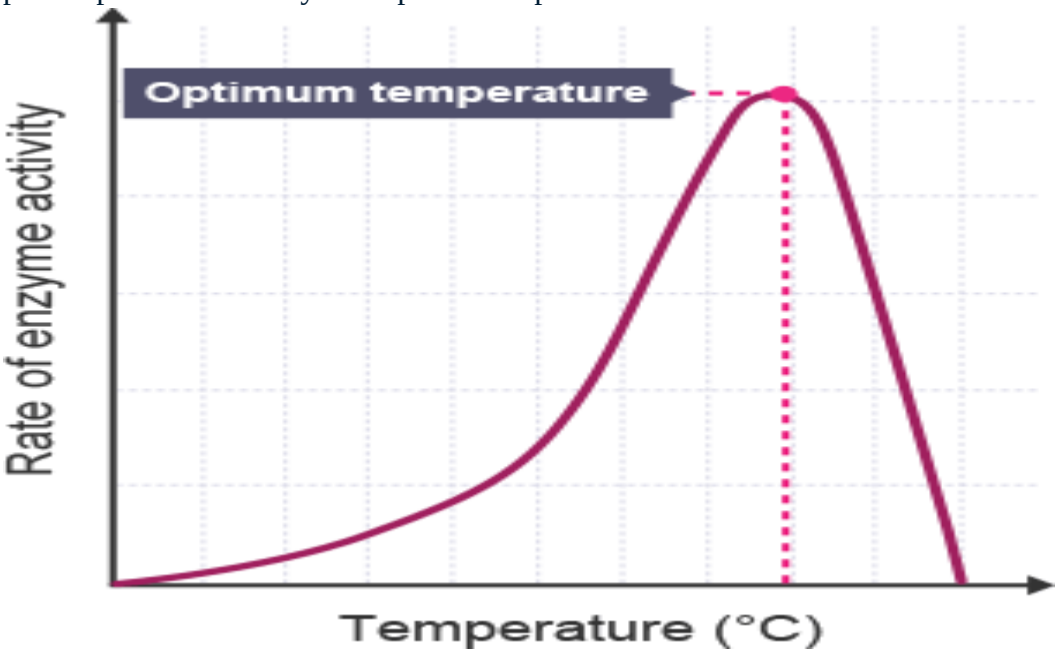
b. OpticalIsomerism -Optical isomerism, a type of stereoisomerism, refers to the phenomenon where molecules with the same molecular formula and bond arrangement have different spatial

arrangements of atoms, resulting in non-superimposable mirror images. These mirror images, called enantiomers, have the same physical properties but differ in their interaction with plane-polarized light.

c. D and L Isomers- D and L isomers, also known as enantiomers, are mirror-image stereoisomers that differ in the arrangement of their chiral carbons. The "D" and "L" designations, derived from Latin ("dexter" and "laevus" meaning right and left), indicate the configuration of the most chiral carbon in a molecule, specifically in carbohydrates and amino acids.

d. Chirality and 2 n -chirality refers to molecules that are non-superimposable mirror images of each other. For a molecule with 'n' chiral centers, the maximum number of possible stereoisomers (including enantiomers and diastereomers) is 2^n.

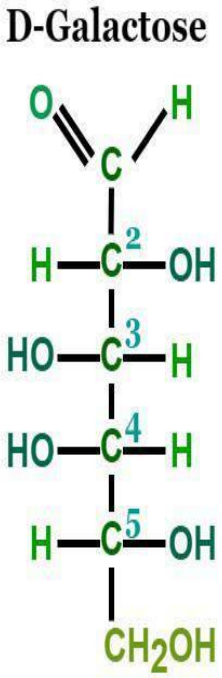
9. **With the help of a graph, explain how temperature affects enzyme activity. [U] / [CO3]** Temperature significantly impacts enzyme activity, with enzyme activity generally increasing as temperature rises until an optimal point, after which it decreases due to denaturation. This relationship can be visualized as a bell-shaped curve where the peak represents the enzyme's optimal temperature.



10. **What are Lipids? Narrate the functions of lipids. [U] / [CO4]**
    Lipids are a diverse group of organic molecules, including fats, oils, waxes, and sterols, that are insoluble in water but soluble in organic solvents. They are a primary source of energy storage and play crucial roles in cell structure, signaling, and various metabolic processes.

Lipids are essential biological molecules with diverse functions, including energy storage, forming cell membranes, and regulating hormones and nerve impulses. They act as a concentrated energy source, cushion organs, and are vital for transporting fat-soluble vitamins. Additionally, lipids are crucial for cell structure, hormone production, and proper brain function.

11. **Elucidate the structure and properties of D-Galactose [U] / [CO1]**
    D-Galactose is a monosaccharide sugar, a simple carbohydrate with the chemical formula C6H12O6, and the molecular mass of 180.156 g/mol. It's known as a C-4 epimer of glucose, meaning the only difference between the two is the position of a hydroxyl group on the fourth carbon atom. Galactose exists in both open-chain and cyclic forms, with the cyclic form having two anomers (alpha and beta)

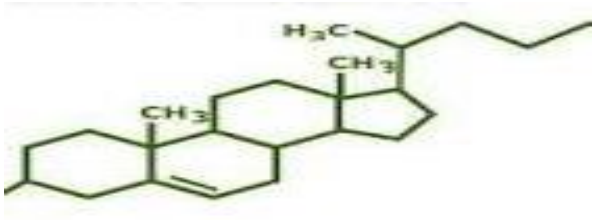


12. Elaborate in detail on the physical properties of amino acids. [U] / [CO2]

Amino acids exhibit several notable physical properties including their crystalline structure, high melting points, variable solubility in water, and amphoteric nature. They are also optically active, meaning they rotate plane-polarized light, and they can exist as zwitterions in solution.

13. With the help of a structure, explain the properties and functions of cholesterol. [U] / [CO4]

Cholesterol is a crucial lipid with a four-ring structure that acts as a building block for cell membranes, steroid hormones, and bile acids. It also helps regulate membrane fluidity and is synthesized within the body, though excessive dietary intake can be detrimental to health.



14. Enumerate the chemical reactions of amino acids due to NH3 group. [K] / [CO2] [4x5 = 20]

The NH3+ group in amino acids participates in various chemical reactions, primarily due to its ability to act as a proton donor or acceptor. These reactions include acid-base interactions, peptide bond formation, and reactions with other functional groups like carboxylic acids. acceptor. These reactionsinclude deamination, transamination, and reactions with various reagents likenitrous acid, aldehydes, and acid chlorides.

Part C

Long Essay Questions

Answer any 2 questions Each question carries 10 marks

15. Explain different types of reversible inhibition occuring in an enzyme catalysed reaction.[A] / [CO3]

There are three main types of reversible inhibitor:

- Competitive inhibitor.
- Non-competitive inhibitor.
- Uncompetitive inhibitor.

16. Describe in detail on classification of carbohydrates. [U] / [CO1]

Carbohydrates are broadly classified into monosaccharides, oligosaccharides, and polysaccharides based on their structural complexity and the number of monosaccharide units they contain. Monosaccharides are the simplest form, acting as building blocks, while oligosaccharides are short chains of 2-10 monosaccharide units, and polysaccharides are long chains of many monosaccharides.

1. Monosaccharides: These are the simplest carbohydrates and cannot be further hydrolyzed into smaller units. **Examples:** Glucose, fructose, galactose.

- **Classification:** Further classified based on the number of carbon atoms (trioses, tetroses, pentoses, hexoses) and the presence of an aldehyde or ketone group (aldoses and ketoses).
- Disaccharides (sucrose, maltose, lactose), trisaccharides (raffinose), tetrasaccharides (stachyose).

3. Polysaccharides: These are long chains of many monosaccharides linked together.

Examples:

Starch (energy storage in plants), glycogen (energy storage in animals), cellulose (structural component in plants), chitin (structural component in fungi and exoskeletons of arthropods).

- **Simple carbohydrates:** Monosaccharides and disaccharides are considered simple carbohydrates due to their ease of digestion.
- **Complex carbohydrates:** Oligosaccharides and polysaccharides are complex carbohydrates due to their larger structure.

17. Write an essay on the structural organisation of proteins. Add a note on the chemical bonds responsible for its formation. [U] / [CO2]

Structural organisation of proteins -Primary, Secondary, Tertiary & quaternary structure

Chemical bonds responsible for its formation-peptide bonds, hydrogen bonds, disulfide bonds, ionic interactions, and hydrophobic interactions.

18. Elaborate in detail on the classification of fatty acids with examples. Make a note on the structure and functions of essential fatty acids.[U] / [CO4]

Fatty acids are classified based on their degree of saturation, length, and whether they are essential or not. Saturated fatty acids have no double bonds, while unsaturated fatty acids have one or more. Essential fatty acids, like omega-3 and omega-6, must be obtained from the diet because the body cannot synthesize them.

[10x2 = 20]